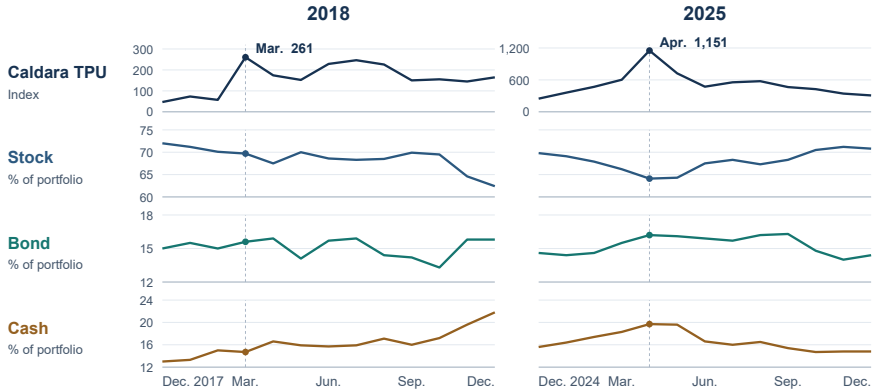
The Impact of Tariff and Trade Policy Uncertainty Shocks on Household Asset Portfolios

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Trade policy uncertainty and portfolios in 2018 and 2025



- The stock share fell into both TPU peaks. Cash and bond shares rose into the April 2025 peak.

Sources: Caldara et al. (2020) and AAII Asset Allocation Survey. TPU scales differ by year. AAII scales are shared.

Main Question

- **How does trade policy uncertainty affect household portfolios?**

Empirical evidence

- Check whether the TPU–equity price link in prior work holds in recent U.S. data.
- Estimate portfolio and real responses to TPU with lagged macro-financial controls.

Theory

- Firms build or draw down import inventories as tariff policy changes.
- Households face ambiguity about inventories and hedge tariff-driven price jumps.

Related literature

- **Trade policy uncertainty.** Caldara et al. (2020)
- **Inventories and trade.** Alessandria, Kaboski, and Midrigan (2011), Khan and Khederlarian (2021)
- **Portfolio choice and ambiguity.** Merton (1971), Drechsler (2013)
- **Nominal wage rigidity.** Christiano, Eichenbaum, and Evans (2005)

Monthly local projections

$$u_t^{\mathrm{TPU}} = \frac{T_t - \widehat{a} - \sum_{\ell=1}^{6} \widehat{\rho}_\ell T_{t-\ell} - \widehat{d}' D_t}{\widehat{\sigma}_\nu},$$

$$W_{t-1} = \left(100\Delta \log P_{t-1}^{\mathrm{PCE}},\ \log \overline{\mathrm{VIX}}_{t-1},\ 100\Delta \log IP_{t-1}\right)',$$

$$y_{t+h} - y_{t-1} = \alpha_h + \beta_h u_t^{\mathrm{TPU}} + \Gamma_h' W_{t-1} + \varepsilon_{t,h}, \qquad h = 0, \ldots, 10.$$

- **Innovation.** T_t is the Caldara et al. (2020) TPU index, D_t contains pandemic indicators, and $\widehat{\sigma}_\nu$ is the residual SD.
- **Prior-month controls.** P^{PCE} is the PCE price index, IP is industrial production, and $\overline{\mathrm{VIX}}$ is average VIX.
- **Sample.** Jan. 2007–Jan. 2026, omitting windows over Mar.–Jun. 2020. Pointwise 95% HAC intervals.

TPU episodes Inference Data sources

Table 1. Portfolio shares

Cumulative responses to a one-standard-deviation TPU innovation

Share	Impact		Month 1		Month 2	
Stock	−0.286	[−0.409, −0.163]	−0.401	[−0.567, −0.235]	−0.269	[−0.482, −0.055]
Cash	0.192	[0.066, 0.319]	0.258	[0.100, 0.416]	0.180	[0.009, 0.350]
Bond	0.096	[0.045, 0.146]	0.146	[0.067, 0.225]	0.093	[0.006, 0.181]

- **Portfolio response.** Stock shares fall while cash and bond shares rise. All three moves peak at month one and partly reverse by month two, with cash gaining more than bonds at each horizon.
- **Model link.** Inventory purchases and drawdowns shift firms' dividend payouts and tariff receipts. Those receipts support government bonds, whose market value can jump at a tariff switch while cash stays continuous.

AAII. Estimates in pp. Pointwise 95% HAC intervals.

Response paths Robustness

Table 2. Market prices and real activity

Outcome	Impact	Month 2
NASDAQ, monthly average	$-\mathbf{0.867}$ $[-1.376, -0.358]$	-0.498 $[-1.226, 0.229]$
Industrial output	0.006 $[-0.038, 0.050]$	0.000 $[-0.052, 0.052]$
Employment	-0.004 $[-0.015, 0.006]$	-0.011 $[-0.039, 0.018]$
Real income	0.011 $[-0.013, 0.035]$	-0.043 $[-0.103, 0.017]$
Real sales	0.016 $[-0.043, 0.075]$	-0.010 $[-0.092, 0.072]$
Real PCE	-0.003 $[-0.028, 0.022]$	-0.033 $[-0.074, 0.008]$

- **Equity revaluation.** The monthly-average NASDAQ falls on impact. In the model, a lower stock price can reduce the stock share at unchanged holdings.
- **Income.** Aggregate real income is imprecisely estimated. The model tracks labor income at a sticky nominal wage.

Cumulative 100 log changes. Pointwise 95% HAC intervals. Controls include lagged monthly-average VIX.

NASDAQ paths Activity paths Aggregate spending Spending components

A two-country portfolio model

- **Two countries in continuous time.** The initial allocation is symmetric in regime 1 before policy news or a tariff switch. Countries trade final goods and intermediates.
- **Production.** Firms use imported inventories and choose employment at a sticky nominal wage. Intermediate supply is infinitely elastic at exogenous prices, and markups allow firms to earn operating profits.
- **Merton households.** Households infer inventories from market signals because firms' inventory decisions are unobserved. Households face uncertainty because they do not know the probability law governing firms' inventory responses to shocks.
- **Partial equilibrium.** To isolate portfolio demand, households take asset price processes as given. Goods and labor markets clear, debt issuance closes the fiscal accounts, and holdings match asset values.
- **Indices.** i is the home country, $j \neq i$ its trading partner, and k, l index policy regimes.

4.1 Trade policy regimes

- Regime 1 represents a period of relative stability in trade policy. Regime 2 represents a period of heightened trade conflict and policy instability, captured in the model by tariff escalation.

$$\mathbf{Q}(t) = \begin{pmatrix} -\lambda_{12}(t) & \lambda_{12}(t) \\ \lambda_{21}(t) & -\lambda_{21}(t) \end{pmatrix}, \qquad \lambda_{12}(t) > 0, \quad \lambda_{21}(t) > 0,$$

$$\tau_{1,2} > \tau_{1,1} = \tau_{2,1} = \tau_{2,2},$$

$$\tau_{i,l} = \text{import tariff in country } i \text{ and regime } l.$$

- $\lambda_{12}(t)$ is the transition rate from regime 1 to 2, and $\lambda_{21}(t)$ is the reverse rate. Only country 1 raises its tariff in regime 2.
- Escalation news raises $\lambda_{12}(t)$ in either regime. In regime 1, this raises the expected tariff on future imports.

Policy events and revaluation

4.2 Production and input sourcing

- Firms price at a markup and hire at a sticky nominal wage. X_i is a CES bundle of domestic ($X_{D,i}$) and foreign ($X_{F,i}$) inputs.

$$Y_i = A_i L_i^{\alpha_i} X_i^{1-\alpha_i},$$

$$X_{F,i} = X_{F,i}^N + X_{F,i}^I,$$

$$P_{F,i}^{\text{use}} = \phi_i P_{M,i} + (1-\phi_i) P_{I,i}, \qquad \phi_i = X_{F,i}^N / X_{F,i}.$$

- $X_{F,i}^N$ is new imports used in production and $X_{F,i}^I$ is withdrawals from inventory. $P_{I,i}$ is the average acquisition cost of stored inputs.
- When the tariff raises the new import price $P_{M,i}$, inventories bought earlier can cushion current production costs.

4.2 Physical inventory and its book value

$$\dot{I}_i = Q_{X,i} - X^I_{F,i} - \delta_I I_i,$$

$$\frac{d(P_{I,i} I_i)}{dt} = P_{M,i} Q_{X,i} - P_{I,i} X^I_{F,i} - \delta_I P_{I,i} I_i.$$

- $Q_{X,i}$ is the flow of newly imported inputs into storage. δ_I measures physical stock loss in storage (inventory depreciation).
- Additions enter at the current import price. Withdrawals and depreciation are recorded at the average acquisition cost.
- A tariff increase changes the cost of new purchases immediately. The inherited stock and its book value carry over at the switch.

Inventory policy

4.2 Inventory finance and dividends

- The firm maximizes discounted operating profit. Storage purchases must be funded from current cash resources $\mathcal{R}_i$.

$$\mathcal{R}_i = \Pi_i + P_{I,i} X^I_{F,i} + \delta_I P_{I,i} I_i,$$
$$0 \leq P_{M,i} Q_{X,i} \leq \mathcal{R}_i,$$
$$CF^S_i = \mathcal{R}_i - P_{M,i} Q_{X,i}.$$

- Π_i is accrued operating profit. Adding back withdrawals and depreciation gives cash available before storage purchases.
- The dividend CF^S_i is the cash left after buying inventory. It is zero when the funding bound binds.

Proposition 1

4.3 Tariff receipts and public debt

$$T_i = \tau_c P_{C,i} C_i + \tau_s P_{Y,i} Y_i + \underbrace{\tau_{i,k} \epsilon_{ij} p_{x,j} (X^N_{F,i} + Q_{X,i})}_{\text{duties on new imports}},$$

$$\dot{D}_i = r^D_{i,k} D_i - (T_i - \bar{G}_i), \qquad G_i(t) \equiv \bar{G}_i > 0.$$

- Duties apply when imports cross the border. ϵ_{ij} converts the partner's currency into country i's currency.
- Nominal government purchases $\bar{G}_i$ are fixed and positive. Net issuance raises D_i when debt service exceeds the primary surplus $T_i - \bar{G}_i$.

Debt service and bond valuation

4.4 Stock, bond, and cash returns

Gains on inherited positions within regime k

$$\frac{dS_{i,t}}{S_{i,t}} = \mu^S_{i,k}\, dt + \sigma^S_{i,k}\, dZ^S_{i,t},$$

$$\frac{dB_{i,t}}{B_{i,t}} = \mu^B_{i,k}\, dt + \sigma^B_{i,k}\, dZ^B_{i,t},$$

$$\frac{dM_{i,t}}{M_{i,t}} = r^M_{i,k}\, dt.$$

- $\mu^S_{i,k}$ and $\mu^B_{i,k}$ are conditional expected total returns, including payouts. $dZ^S_{i,t}$ and $dZ^B_{i,t}$ are correlated return innovations.
- Equity pays cash profits and bonds pay debt service, and both prices can jump at a regime switch. Cash accrues the safe rate and remains continuous.

Valuation and net issuance

4.4 Incomplete information about inventories

- Firms know their inventories. Households update a conditional distribution using observed prices, returns, wages, payouts, and the public policy regime.

$$dI_{i,t} = b_{i,k}(I_{i,t})\,dt + \bar{\nu}_{i,k}\,dW^I_{i,t}, \qquad \bar{\nu}_{i,2} > \bar{\nu}_{i,1}.$$

- In the household's reference forecast, $b_{i,k}$ is net accumulation implied by the sourcing rule and $\bar{\nu}_{i,k}$ is the dispersion of inventory innovations.

$$\Psi^i_k = \psi^i_k(I)\,\bar{\nu}_k,$$

$$\sigma^i_k = \sqrt{[R + \boldsymbol{\Psi}_k \boldsymbol{\Psi}_k^\top]_{ii}} = \sqrt{\Sigma_{k,ii}}, \qquad i \in \{S, B\}.$$

- $\psi^i_k(I)$ is the sensitivity of log asset value to inventory. σ^i_k is total return volatility, with R capturing covariance orthogonal to inventory shocks.

4.4 Ambiguity about inventory dynamics

- Households consider alternative inventory drifts around the reference forecast.

$$dI_{i,t} = [b_{i,k}(I_{i,t}) + \bar{\nu}_{i,k} h_{i,t}]\, dt + \bar{\nu}_{i,k}\, dW_{i,t}^{I,\mathbb{Q}_i}, \qquad \tfrac{1}{2} h_{i,t}^2 \le \kappa_k(t),$$

$$\kappa_k(t) = \bar{\kappa}\, \mathbf{1}_{\{k=2\}} + \varphi_\kappa \frac{\lambda_{12}(t)}{\lambda^{\max}}.$$

- The distortion $h_{i,t}$ changes the drift at a given diffusion. The entropy bound κ_k widens with escalation news and has a higher regime component in regime 2.
- With positive total inventory exposure, the worst case shifts the drift down. At fixed positive loadings, equity takes the larger return reduction under the model's loading assumption.
- Relative-entropy formulation following Drechsler (2013).

Worst-case distortion

4.5 Robust Merton portfolio choice

$$J_{i,k} = \max_{C_i, \boldsymbol{\theta}_i} \min_{\mathbb{Q}_i \in \mathcal{P}_{i,k}} \mathbb{E}^{\mathbb{Q}_i}_{i,0} \int_0^\infty e^{-\rho t} \left[\frac{C_{i,t}^{1-\gamma}}{1-\gamma} - \frac{l_{i,t}^{1+\chi}}{1+\chi} \right] dt,$$

$$dW_{i,t} = \left[W_{i,t}(r^M_{i,k} + \boldsymbol{\theta}_{i,t}^\top \boldsymbol{\pi}_{i,k}) - \widetilde{P}_{C,i,t} C_{i,t} + w_{i,t} l_{i,t} \right] dt + W_{i,t} \boldsymbol{\theta}_{i,t}^\top \boldsymbol{\sigma}_{i,k} \, d\boldsymbol{Z}_{i,t},$$

$$\theta^S_{i,t}, \theta^B_{i,t} \geq 0, \quad \theta^S_{i,t} + \theta^B_{i,t} \leq 1, \quad \theta^M_{i,t} = 1 - \theta^S_{i,t} - \theta^B_{i,t}.$$

- $\boldsymbol{\theta}_{i,t}$ gives risky shares, and $\boldsymbol{\pi}_{i,k}$ gives their expected excess returns. $\mathcal{P}_{i,k}$ is the set of admissible probability laws.
- The budget governs wealth between switches, which reprice inherited risky holdings. Employment follows firms' labor demand, and wealth and consumption stay nonnegative.

4.6 Market clearing and external balance

$$L_i = l_i, \qquad P_{Y,i} Y_i = P_{C,i} C_i + \bar{G}_i + NX_i^F,$$

$$NX_1^F(t) = \mathcal{M}_1(t) - \mathcal{M}_2(t).$$

- **Goods and labor.** NX_i^F is final-goods net exports in country i's currency. Firms choose labor at the posted wage.
- **Intermediate supply.** Exporters fill all orders at $p_{x,j}$ in their currency. The exchange rate converts that price into the importer's currency.
- **Trade payments.** $\mathcal{M}_i$ is country i's pretariff import bill, including stockbuilding, in country 1's currency. With no foreign borrowing or reserve flows, net final-goods exports pay for net intermediate imports.
- **Exchange rate.** It adjusts with output and final spending to balance bilateral payments.

Proposition 3

Proposition 2. Optimal inventory policy

$$\lambda_{12}(P_{M,i,2} - P_{M,i,1}) > (\rho + \delta_I)P_{M,i,1}, \qquad \bar{q}_i = \frac{\lambda_{21}P_{M,i,1}}{\rho + \delta_I + \lambda_{21}},$$

$$T_i^* = \frac{1}{\rho + \delta_I + \lambda_{21}} \log\left[\frac{P_{M,i,2} - \bar{q}_i}{\left(1 + \frac{\rho+\delta_I}{\lambda_{12}}\right) P_{M,i,1} - \bar{q}_i}\right].$$

- **Positive target.** In the auxiliary problem, the escalation intensity times the import price gap must exceed the cost of carrying one more unit in regime 1.
- **Drawdown.** Starting at the regime 1 target, T_i^* is the time to exhaust the stock in regime 2.
- **Purchases over time.** The cash funding limit can spread purchases over time. An early switch leaves a smaller stock.

Stock target $I_{i,1}^*$ Funding limit

Proposition 5. Cash flows and equilibrium values

$$\dot{M}_i = r^M_{i,k} M_i + w_i l_i - (1+\tau_c) P_{C,i} C_i + CF^S_i + T_i - \bar{G}_i.$$

$$(S_i(t), B_i(t), M_i(t)) = \lim_{T\to\infty} \mathbb{E}^{h^*_i}_{i,t} \int_t^T \frac{m_{i,s}}{m_{i,t}} \left(CF^S_i(s), T_i(s) - \bar{G}_i, K_i(s) \right) ds,$$

$$W_i(t) = \lim_{T\to\infty} \mathbb{E}^{h^*_i}_{i,t} \int_t^T \frac{m_{i,s}}{m_{i,t}} \left[(1+\tau_c) P_{C,i}(s) C_i(s) - w_i(s) L_i(s) \right] ds.$$

- **Cash today.** Wages and investment income fund spending or add to cash. Net bond receipts equal coupons less net issuance.
- **Asset values.** Discounted dividends back equity, and primary surpluses back bonds. The cash-account flow $K_i = r^M_{i,k} M_i - \dot{M}_i$ reflects intermediate payments.
- **Financial wealth.** It funds future consumption spending net of labor income.

Marginal utility and valuation

Proposition 6. Portfolio responses to policy events

- **Policy event.** Inherited cash stays fixed while equity and bonds can reprice. At marked wealth, households reoptimize under the updated policy law.

$$W_i^+ = W_i^- \left(\theta_i^{S,-} \frac{S_i^+}{S_i^-} + \theta_i^{B,-} \frac{B_i^+}{B_i^-} + \theta_i^{M,-} \right),$$

$$\mathrm{d}\theta_i^M > 0, \quad \mathrm{d}\theta_i^S < 0 \quad \text{if both risky prices fall and equity falls by a larger percentage,}$$

$$\mathrm{d}\theta_i^B \gtrless 0 \quad \Longleftrightarrow \quad \mathrm{d}B_i \gtrless \theta_i^{B,-}\, \mathrm{d}W_i.$$

- **Equity and cash.** With positive inherited holdings, equity loses more in percent than wealth. The fixed cash balance gains share as wealth falls.
- **Bonds.** The bond share rises if bonds lose less than wealth in percent, and falls if they lose more. Cash gains more than a rising bond share if the stock loss in share exceeds twice the bond gain.

Comparative-statics conditions Covariance and hedging

Proposition 7. Hedging a tariff switch

- **Why hedge?** Firms' inventories shape how dividends and debt service change at a tariff switch. Households can adjust their portfolios ahead of the resulting price jumps.

$$\text{Optimal risky shares} = \text{Shares without switch hedging} + \text{Switch hedge}.$$

- **Without switch hedging.** Households weigh risk against returns under the worst-case inventory forecast. When household value increases with inventory, that forecast lowers expected returns on assets that benefit from larger stocks of inputs.
- **Switch hedge.** The hedge compares how the two assets move at the switch with how they usually move together. If they usually move together and bonds lose sufficiently less at the switch, the hedge can reduce stock demand and raise bond demand.

Full portfolio rule When the bond hedge is positive

Conclusion

- **Evidence.** Impact shares (pp): equity -0.286, cash $+0.192$, bonds $+0.096$.
- **Inventories.** The auxiliary inventory target can be positive in regime 1 and is zero in regime 2. Buying early and drawing stocks down after escalation change equity payouts and the timing of tariff receipts.
- **Ambiguity.** Regime 2 has greater inventory dispersion and a wider set of drift laws. Escalation news can lower ambiguity-adjusted returns before tariffs change and strengthen incentives to hedge revaluation at a switch.
- **Contribution.** Inventories connect firm dividends and public debt service to household hedging. Equity can fall and cash can rise on impact, while the bond share depends on its revaluation relative to wealth.

Thank you

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Conclusion Appendix

Appendix

Empirical evidence

TPU index Portfolio paths NASDAQ Activity Aggregate PCE

Table 3 Spending components Table 5 Table 6 Table 7

Inference Table A.1. Data

Production and market clearing

Table 8 Proposition 1 Stock target Funding time Proposition 3

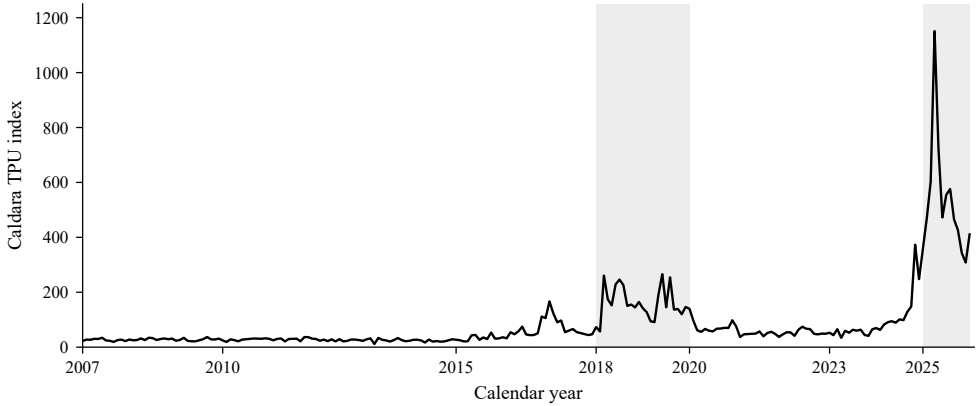
Households and asset prices

Proposition 4 Bellman equation Asset valuation Equation (62)

Proposition 6 conditions Proposition 7 rule Hedge conditions

Conclusion Thank you

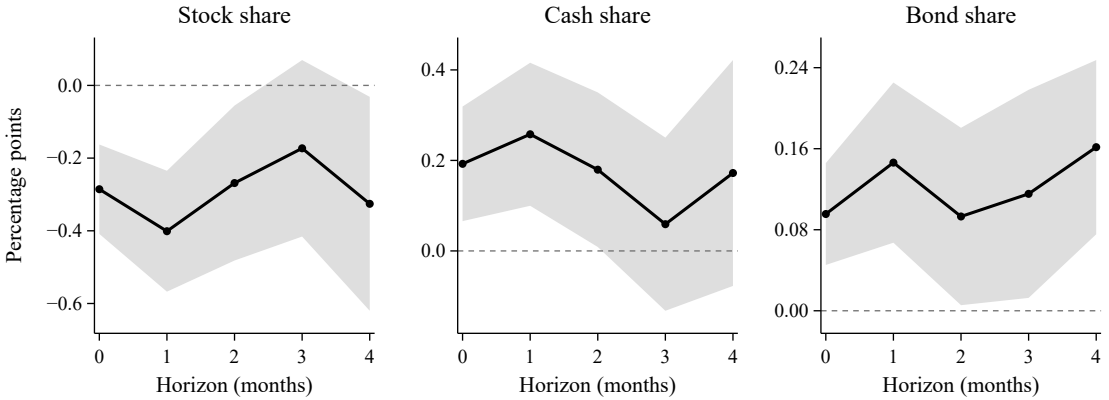
Caldara trade policy uncertainty index



- **Major episodes.** Shading marks the 2018–19 trade conflict and the 2025 escalation. The index peaks at 266 in June 2019 and 1,151 in April 2025. The first stage uses published index levels.

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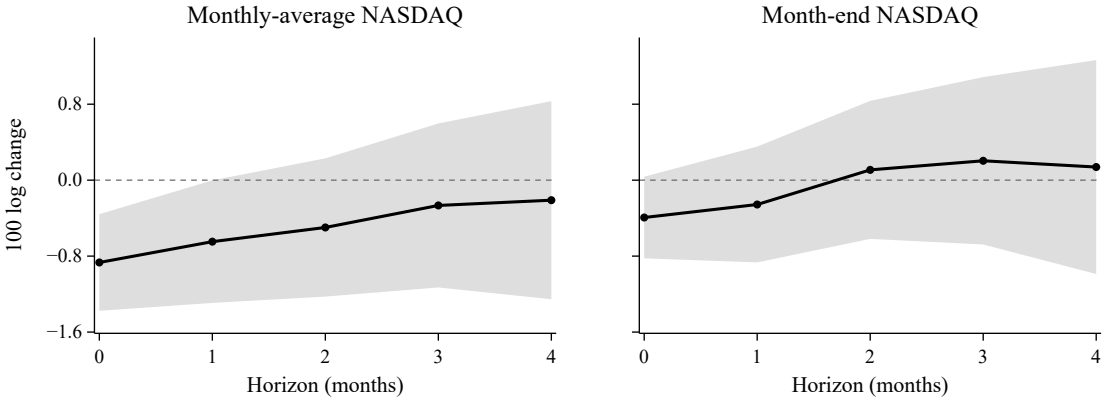
Portfolio share response paths



- **Portfolio weights.** The paths show cumulative changes in percentage points. Panels have separate vertical scales. Shading shows pointwise 95% HAC intervals.

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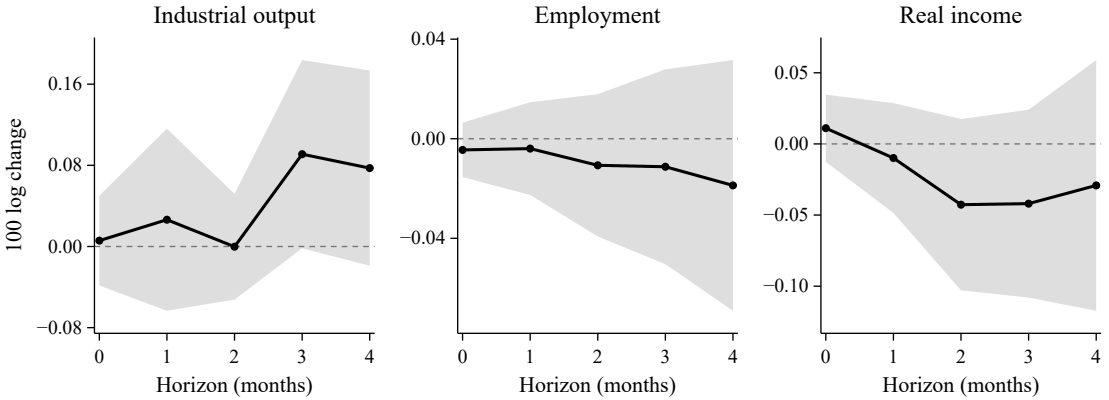
NASDAQ response by measurement timing



- **Price timing.** The monthly-average NASDAQ uses daily closes and the month-end series uses the last close. Both regressions control for lagged monthly-average VIX.

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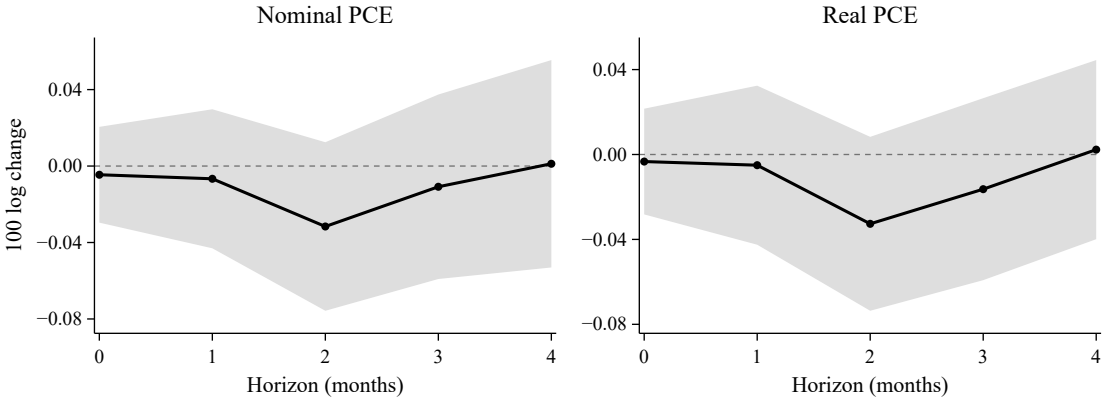
Output, employment, and real income



- **Real activity.** The paths show cumulative 100 log changes. Panels have separate vertical scales. Shading shows pointwise 95% HAC intervals.

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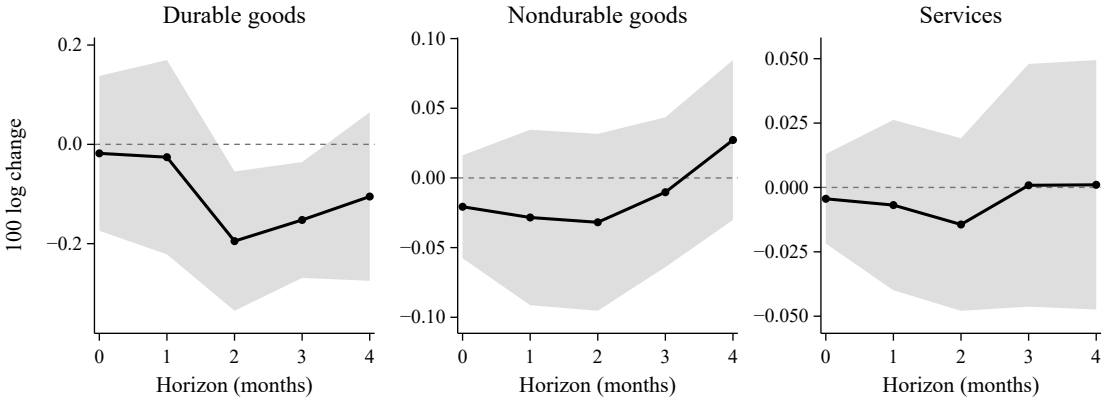
Nominal and real spending



- **Spending.** The paths show cumulative 100 log changes. Nominal PCE measures spending at current prices. Real PCE measures the volume of expenditure.

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Real spending by category



- **Durable versus nondurable spending.** The comparison uses common regression dates. Durable spending falls at months two and three, with pointwise intervals excluding zero.

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Inference and the April 2025 innovation

- **Concentration.** At impact, 2025 accounts for 70% of the squared residualized TPU innovation. April 2025 alone accounts for 44%.

Share (pp)	Baseline	Drop Apr. 2025	95% CI after drop
Stock	-0.286	-0.328	$[-0.549, -0.106]$
Cash	0.192	0.215	$[-0.014, 0.443]$
Bond	0.096	0.117	$[0.032, 0.203]$

- The first-stage fit and scale stay fixed. The impact signs survive the deletion and the cash interval includes zero.
- All three impact intervals exclude zero under all seven covariance estimators. At month two, this holds for three of seven per share. Intervals are pointwise and conditional on the fitted first stage.

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Table A.1. Data sources and measurement

Series	Source and measurement
Portfolio shares	AAII monthly stock, bond, and cash weights.
Trade uncertainty	Caldara et al. (2020) TPU in published level units.
Consumer prices	BEA/FRED PCEPI. Monthly log inflation.
Market risk	Cboe/FRED VIXCLS. Log of the previous month's average daily close.
Equity prices	Nasdaq/FRED NASDAQCOM. Monthly average and last daily close.
Industrial output	Federal Reserve/FRED INDPRO. Seasonally adjusted.
Real consumption	BEA/FRED PCEC96. Real expenditure in chained dollars.

More series Back to main slide Appendix menu

Table A.1. Data sources continued

Series	Source and measurement
Nominal spending	BEA/FRED PCE. Current-dollar expenditure.
Other uncertainty	Baker, Bloom, and Davis general and trade EPU.
Other outcomes	Employment, income, sales, and intermediate-demand PPI in cumulative log changes.
Spending components	BEA/FRED PCEDGC96, PCENDC96, and PCESC96.
Nominal wage growth	Atlanta Fed/FRED FRBATLWGTUMHWGO. Median year-over-year hourly wage growth.
Bond yield	FRED DGS10. Yield changes with an eight-year duration for risk moments.

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Table 3. Total and durable spending

Real spending category	h	Estimate	95% HAC interval	p	N
Total real PCE	0	−0.009	[-0.031, 0.012]	0.407	223
	1	−0.014	[-0.058, 0.031]	0.547	221
	2	−0.038	[-0.080, 0.004]	0.076	219
	3	−0.018	[-0.061, 0.025]	0.416	217
	4	−0.005	[-0.048, 0.038]	0.829	215
Durable goods	0	−0.018	[-0.174, 0.138]	0.819	223
	1	−0.026	[-0.221, 0.170]	0.795	221
	2	−0.195	[-0.335, -0.055]	0.007	219
	3	−0.152	[-0.269, -0.036]	0.011	217
	4	−0.105	[-0.275, 0.065]	0.224	215

- **Durable spending.** The pointwise intervals exclude zero for declines at months two and three. Estimates use common regression dates. Units are cumulative 100 log changes.

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Table 3. Nondurable spending and services

Real spending category	h	Estimate	95% HAC interval	p	N
Nondurable goods	0	−0.021	[-0.058, 0.016]	0.271	223
	1	−0.028	[-0.091, 0.035]	0.375	221
	2	−0.032	[-0.095, 0.032]	0.324	219
	3	−0.010	[-0.064, 0.043]	0.707	217
	4	0.027	[-0.030, 0.085]	0.351	215
Services	0	−0.004	[-0.022, 0.013]	0.618	223
	1	−0.007	[-0.040, 0.026]	0.683	221
	2	−0.014	[-0.048, 0.019]	0.398	219
	3	0.001	[-0.046, 0.048]	0.972	217
	4	0.001	[-0.047, 0.049]	0.967	215

- **Nondurables and services.** Pointwise 95% HAC intervals include zero at the five reported horizons. The component series share the October 23, 2026 data vintage.

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Table 5. Alternative sample starts

Sample start	Stock	Cash	Bond	N
January 2007	−0.286	0.192	0.096	224
January 2010	−0.279	0.180	0.101	188
January 2015	−0.294	0.190	0.105	128

- **Common scale.** Impact estimates are in percentage points. The first-stage coefficients and scaling are fixed. All samples end in January 2026 and retain the two major tariff episodes.

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Table 6. Alternative uncertainty measures

Indicator	Stock	Cash	Bond
Baseline TPU	$-0.286\,[-0.409, -0.163]$	$0.192\,[0.066, 0.319]$	$0.096\,[0.045, 0.146]$
U.S. EPU	$-0.594\,[-0.915, -0.272]$	$0.356\,[-0.008, 0.720]$	$0.240\,[-0.038, 0.519]$
Trade EPU	$-0.280\,[-0.627, 0.067]$	$0.206\,[-0.105, 0.517]$	$0.075\,[-0.198, 0.348]$

- **Alternative indices.** Estimates are in percentage points per one standard deviation of each innovation. General EPU reproduces the stock decline. The trade EPU intervals are wider. Both alternatives use AR(6) residuals from log indices.

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Table 7. Additional lagged controls

Added control	Stock	Cash	Bond
Baseline	$-0.286\,[-0.409, -0.163]$	$0.192\,[0.066, 0.319]$	$0.096\,[0.045, 0.146]$
policy rate	$-0.286\,[-0.412, -0.160]$	$0.194\,[0.063, 0.324]$	$0.095\,[0.042, 0.148]$
market return	$-0.151\,[-0.285, -0.017]$	$0.054\,[-0.088, 0.196]$	$0.100\,[0.030, 0.171]$
past stock share	$-0.210\,[-0.374, -0.047]$	$0.137\,[-0.023, 0.298]$	$0.075\,[0.024, 0.127]$

- **Sensitivity to controls.** Each added control enters at a one-month lag ($N = 224$ throughout). With market returns or past stock shares, the stock and bond intervals exclude zero and the cash interval includes zero.

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Table 8. Preset structural parameters

Parameter	Meaning	Value	Reference
γ	Risk aversion	5	Campbell et al. (2002)
ρ	Discount rate	0.04/yr	
χ	Labor disutility curvature	1	Chetty et al. (2011)
α_i	Labor share	0.62	Gollin (2002)
η	Input substitution elasticity	2	Boehm et al. (2023)
ε_p	Variety elasticity	6	Christiano et al. (2005)
s, s_c	Domestic CES weights	0.85	Amiti et al. (2019)
δ_I	Inventory depreciation	0.25/yr	Alessandria et al. (2011)
τ_c	Consumption tax	0.065	
τ_s	Sales tax	0.05	
$\tau_{i,1}$	Common baseline tariff	0.03	Fajgelbaum et al. (2020)
$\tau_{1,2}$	Country 1 escalation tariff	0.12	Fajgelbaum et al. (2020)

- **Calibration inputs.** References motivate the parameter choices. Values are maintained model inputs.

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Proposition 1. Production and inventory orders

$$P^*_{F,i} = \min\{P_{M,i}, U_i/\Lambda_i\}, \qquad Q_{X,i} \in \begin{cases} \{\mathcal{R}_i/P_{M,i}\}, & \chi_{Q,i} > 0, \\ [0, \mathcal{R}_i/P_{M,i}], & \chi_{Q,i} = 0, \\ \{0\}, & \chi_{Q,i} < 0. \end{cases}$$

- **Definitions.** $Z_i = P_{I,i} I_i$ is book stock. For firm value V_i, let $\chi_{Q,i} = V_{i,I} + P_{M,i} V_{i,Z}$, $\Lambda_i = 1 + (\chi_{Q,i})_+ / P_{M,i}$, and $U_i = V_{i,I} + P_{I,i}(1 + V_{i,Z})$.
- **Sourcing and funding.** Use the cheaper source, or import at stockout. Evaluate markup and factor demands at $P^*_{F,i}$. A positive marginal order value makes the cash cap bind.

$$CF^S_i = \Pi_i - \dot{Z}_i, \qquad \dot{I}_i = X^N_{F,i} + Q_{X,i} - X_{F,i} - \delta_I I_i.$$

- With stationary prescribed inputs and zero inherited stock, accumulation improves operating value exactly when $\lambda_{12}(P_{M,i,2} - P_{M,i,1}) > \delta_I P_{M,i,1}$.

Funding time Back to main slide Appendix menu

Auxiliary stock target and drawdown time

$$T_i^* = \frac{1}{\rho + \delta_I + \lambda_{21}} \log\left[\frac{P_{M,i,2} - \bar{q}_i}{\left(1 + \frac{\rho+\delta_I}{\lambda_{12}}\right) P_{M,i,1} - \bar{q}_i}\right], \qquad \bar{q}_i = \frac{\lambda_{21} P_{M,i,1}}{\rho + \delta_I + \lambda_{21}},$$

$$I_{i,1}^* = \frac{\bar{X}_{F,i}}{\delta_I}\left(e^{\delta_I T_i^*} - 1\right).$$

- **Drawdown.** At a positive auxiliary target, T_i^* is the time to exhaust the inherited stock in regime 2. The constant $\bar{X}_{F,i}$ is foreign input use during drawdown.
- **Stock target.** The low-regime target is $I_{i,1}^*$. With zero depreciation it equals $\bar{X}_{F,i} T_i^*$. The excess target over the benchmark stock is $I_{i,1}^* - K_{0,i}$.

Funding time Back to main slide Appendix menu

Time to build inventory under the funding cap

$$I_i(t) = \min\{I_i^{\mathrm{tar}},\, I_i^{\mathrm{cap}}(1 - e^{-\delta_I t})\}, \qquad I_i^{\mathrm{cap}} = \frac{\pi_{f,i,1}}{\delta_I P_{M,i,1}},$$

$$t_i^{\mathrm{hit}} = -\frac{1}{\delta_I} \log\left(1 - \frac{I_i^{\mathrm{tar}}}{I_i^{\mathrm{cap}}}\right) \qquad (0 < I_i^{\mathrm{tar}} < I_i^{\mathrm{cap}}).$$

- **Funding path.** Starting from zero inventory with $\delta_I > 0$, $\pi_{f,i,1}$ is the maximum pre-depreciation cash surplus at the fresh-input price. I_i^{tar} is the single target of the funded firm under the stationary benchmark.
- **Timing.** Purchases build the stock toward the target over time. If the tariff switches before t_i^{hit}, the firm carries only $I_i(t)$ into the new regime. Targets at or above I_i^{cap} have no finite hitting time.

Order rule Auxiliary drawdown Appendix menu

Proposition 3. Prices and quantities

$$R_{Y,a} = \omega_{a1} E_1 + \omega_{a2} E_2, \qquad Y_1 = \frac{R_{Y,1}}{P_{Y,1}}, \qquad Y_2 = \frac{R_{Y,2}}{eP_{Y,2}},$$

$$\frac{E_2}{E_1} = \frac{\nu_1 + (1 - \nu_1 - \nu_2)\omega_{21}}{\nu_2 + (1 - \nu_1 - \nu_2)\omega_{12}}.$$

- **Final demand.** At the trial exchange rate e, E_b is total final expenditure in destination b, in country 1's currency. ω_{ab} is the CES expenditure share on goods from origin a.
- **External balance.** ν_b is pretariff intermediate imports per unit of producer revenue. The second line chooses e so the final-goods balance offsets the intermediate-goods balance.
- **Conditional uniqueness.** Given positive $J_{b,k,W}$ and fixed firm continuations, the sufficient bounds are $s_c \geq 1/2$, $\alpha_1 + \alpha_2 > 1$, and $\nu_1^{\max} + \nu_2^{\max} < 1$. Here $\nu_b^{\max}$ bounds the import ratio.

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Asset valuation and bond issuance

$$S_i(t) = \mathbb{E}^{\mathrm{pr}}_{i,t} \int_t^\infty A^S_i(t,s)\, CF^S_i(s)\, ds,$$

$$B_i(t) = \mathbb{E}^{\mathrm{pr}}_{i,t} \int_t^\infty A^B_i(t,s)\, r^D_i(s) D_i(s)\, ds.$$

- **Market values.** $A^a_i(t,s) = \exp(-\int_t^s d^a_i(u)\, du)$ discounts gross payouts at the pricing rate d^a_i. Equity pays cash dividends and bonds pay debt service.

$$r^D_{i,k} D_i - \dot{D}_i = T_i - \bar{G}_i,$$

$$\mu^B_{i,k,\mathrm{hold}} = \mu^B_{i,k,\mathrm{gross}} - \mathbb{E}_{i,t}[\dot{D}_i / B_i].$$

- **Household bond receipts.** The domestic household sector absorbs net issuance. Subtracting issuance from gross debt service gives the primary surplus $T_i - \bar{G}_i$ in equation (62).

Equation (62) Asset returns Appendix menu

Proposition 4. What determines optimal risky shares?

$$\boldsymbol{A}_{i,k}\boldsymbol{\theta}_i = \boldsymbol{m}_{i,k} - \sqrt{2\kappa_k}\, s_{i,k}\boldsymbol{\Psi}_{i,k} + \frac{\lambda_{kl}}{J_{i,k,W}}\mathbb{E}_{i,t}^{h^*_{i,k}}[J_{i,l,W}\boldsymbol{\Delta}_{i,kl} \mid k \to l] - \boldsymbol{n}_{i,k}.$$

$$\boldsymbol{A}_{i,k} = -\frac{W_i J_{i,k,WW}}{J_{i,k,W}}\Sigma_{i,k}, \qquad \boldsymbol{m}_{i,k} = \boldsymbol{\pi}^0_{i,k} + \Gamma_{i,k}\frac{J_{i,k,Wz}}{J_{i,k,W}}.$$

- **Portfolio choice.** The condition determines the equity and bond shares jointly. Cash is the remaining share.
- **Risk and learning.** $\boldsymbol{A}_{i,k}$ prices covariance risk using wealth curvature. $\boldsymbol{m}_{i,k}$ adds a hedge against changes in inventory beliefs to reference expected excess returns.
- **Ambiguity and transition hedging.** Inventory ambiguity adjusts expected returns. The jump term values revaluation using post-switch marginal utility, and $\boldsymbol{n}_{i,k}$ accounts for portfolio constraints.

Bellman equation and distortion

Bellman equation and worst-case drift

$$\rho J_{i,k} = \max_{C_i, \boldsymbol{\theta}_i} \min_{|h_i| \le \sqrt{2\kappa_k}} \big\{ U(C_i) - v(l_i) + J_{i,k,t} + (\mathcal{L}^{C_i, \boldsymbol{\theta}_i, h_i} J_i)_k \big\}.$$

- **Continuation.** The generator includes the regime jump at marked wealth. $\upsilon_{i,k}$ measures how h_i changes the drift of beliefs z_i.

$$\mathcal{E}_{i,k} = W_i J_{i,k,W} \boldsymbol{\theta}_i^\top \boldsymbol{\Psi}_{i,k} + \upsilon_{i,k}^\top J_{i,k,z},$$
$$h_{i,k}^* = -\sqrt{2\kappa_k}\, s_{i,k}, \qquad s_{i,k} \in \partial |\mathcal{E}_{i,k}|,$$
$$C_i = [(1+\tau_c) P_{C,i} J_{i,k,W}]^{-1/\gamma}.$$

- **Worst case.** $\mathcal{E}_{i,k}$ combines exposure through financial wealth and through beliefs. Positive exposure selects a downward drift. The consumption rule holds at an interior optimum.

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Household valuation in partial equilibrium

$$m_{i,t} = \frac{e^{-\rho t} C_{i,t}^{-\gamma}}{(1+\tau_c) P_{C,i,t}}, \qquad K_i = r_{i,k}^M M_i - \dot{M}_i.$$

Valuation along the clearing path

$$(S_i(t), B_i(t), M_i(t)) = \lim_{T\to\infty} \mathbb{E}_{i,t}^{h_i^*} \int_t^T \frac{m_{i,s}}{m_{i,t}} \left(CF_i^S(s),\, T_i(s) - \bar{G}_i,\, K_i(s) \right) ds,$$

$$W_i(t) = \lim_{T\to\infty} \mathbb{E}_{i,t}^{h_i^*} \int_t^T \frac{m_{i,s}}{m_{i,t}} \left[(1+\tau_c) P_{C,i}(s) C_i(s) - w_i(s) L_i(s) \right] ds.$$

- **Assets.** The first line prices dividends, the primary surplus, and the intermediate payment K_i.
- **Financial wealth.** The second line prices consumption net of labor income. Both use $m_{i,s}/m_{i,t}$ under the worst-case law at a regular interior optimum.

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Proposition 6. Responses to policy events

$$W_i^+ = W_i^- \left(\theta_i^{S,-} \frac{S_i^+}{S_i^-} + \theta_i^{B,-} \frac{B_i^+}{B_i^-} + \theta_i^{M,-} \right).$$

- **Mark inherited holdings.** Physical and book inventory, nominal debt, and cash carry through the event. The superscripts $-$ and $+$ denote the baseline and updated states.

$$\boldsymbol{A}_i^+ \, \mathrm{d}\boldsymbol{\theta}_i = \boldsymbol{F}_i, \qquad \boldsymbol{F}_i = \mathrm{d}\boldsymbol{b}_i - (\boldsymbol{A}_i^+ - \boldsymbol{A}_i^-)\boldsymbol{\theta}_i^- .$$

- **Reoptimize.** $\boldsymbol{b}_i$ is the right-hand side of the Proposition 4 condition. d denotes the updated value less the baseline value.

$$\mathrm{d}\theta_i^B > 0 \iff (0,1)(\boldsymbol{A}_i^+)^{-1}\boldsymbol{F}_i > 0,$$
$$\mathrm{d}\theta_i^M > \mathrm{d}\theta_i^B \iff (1,2)(\boldsymbol{A}_i^+)^{-1}\boldsymbol{F}_i < 0.$$

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Proposition 7. Full portfolio rule

$$\boldsymbol{x}^{*}_{i,k} - \boldsymbol{x}^{\mathrm{nh}}_{i,k} = \frac{\lambda_{kl}(A_{i,l}/A_{i,k})}{\gamma(1 + \boldsymbol{x}^{*\top}_{i,k}\boldsymbol{\Delta}_{i,kl})^{\gamma}}\Sigma^{-1}_{i,k}\boldsymbol{\Delta}_{i,kl},$$

$$\boldsymbol{x}^{\mathrm{nh}}_{i,k} = \gamma^{-1}\Sigma^{-1}_{i,k}\left(\boldsymbol{\pi}^{0}_{i,k} - \sqrt{2\kappa_k}\boldsymbol{\Psi}_{i,k}\right).$$

- **Shares.** $\boldsymbol{x}^{*}_{i,k}$ and $\boldsymbol{x}^{\mathrm{nh}}_{i,k}$ are stock and bond shares in total-wealth units, with and without the transition hedge. The comparison holds expected returns and covariance fixed.
- **Hedge.** $\boldsymbol{\Delta}_{i,kl}$ gives the proportional price jumps and $\Sigma_{i,k}$ is the return covariance. The positive multiplier includes the transition intensity and the value coefficients $A_{i,k}$ and $A_{i,l}$.
- **Expected returns.** $\boldsymbol{\Psi}_{i,k}$ stacks the stock and bond inventory loadings. With positive total exposure, the worst-case drift lowers expected returns on assets with positive loadings.

Covariance conditions Back to main slide Appendix menu

Covariance conditions for the hedge

$$\Sigma_{i,k}^{-1}\boldsymbol{\Delta}_{i,kl} = \frac{1}{\det \Sigma_{i,k}} \begin{pmatrix} (\sigma_{i,k}^{B})^2 \Delta_{i,kl}^{S} - \sigma_{i,k}^{SB} \Delta_{i,kl}^{B} \\ (\sigma_{i,k}^{S})^2 \Delta_{i,kl}^{B} - \sigma_{i,k}^{SB} \Delta_{i,kl}^{S} \end{pmatrix}.$$

- **Hedge direction.** With two negative price jumps and a positive hedge multiplier, equity demand falls if $(\sigma_{i,k}^{B})^2 |\Delta_{i,kl}^{S}| > \sigma_{i,k}^{SB} |\Delta_{i,kl}^{B}|$. Bond demand rises if $\sigma_{i,k}^{SB} |\Delta_{i,kl}^{S}| > (\sigma_{i,k}^{S})^2 |\Delta_{i,kl}^{B}|$.
- **Inventory exposure.** $\boldsymbol{\Psi}_{i,k}$ affects worst-case expected returns and the covariance matrix. $\boldsymbol{\Delta}_{i,kl}$ records the price changes at the switch.
- **Cash.** Cash rises more than bonds when the decline in the stock share exceeds twice the increase in the bond share.

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